# Year 13 Further Maths - Matrices

Similar questions - Algebra, inverses and solving equations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

**1**

**(a)** The matrices $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ a & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 5 & 4 \\ 6 & 9 \end{pmatrix}$ satisfy $\mathbf{AB} = \mathbf{BA}$. Find the value of $a$. **[3]**

**(b)** $\mathbf{C} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 0 & 2 & 3 \end{pmatrix}$. Find **CD** and **DC**. **[3]**

**(c)** State the property of matrix multiplication illustrated by $\mathbf{CD} \neq \mathbf{DC}$. **[1]**

**2** $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 4 & -2 \end{pmatrix}$.

**(a)** Find $3\mathbf{A} - 2\mathbf{B}$. **[2]**

**(b)** Find $\det \mathbf{A}$ and write down $\mathbf{A}^{-1}$. **[3]**

**(c)** Hence solve $3x + 2y = 8$, $-x + 4y = 2$. **[2]**

**(d)** Write down a matrix **N** such that $\mathbf{AN} = 14\mathbf{I}$. **[1]**

**Question 1**

(a) $\mathbf{AB} = \begin{pmatrix} 17 & 22 \\ 5a+18 & 4a+27 \end{pmatrix}$ and $\mathbf{BA} = \begin{pmatrix} 5+4a & 22 \\ 6+9a & 39 \end{pmatrix}$. Equating: $17 = 5+4a$ gives $a = 3$ (check: $5a+18 = 33 = 6+27$ and $4a+27 = 39$)

(b) $\mathbf{CD} = \begin{pmatrix} 0 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 4 & 6 \end{pmatrix}$ and $\mathbf{DC} = (6)$

(c) Matrix multiplication is not commutative

**Question 2**

(a) $\begin{pmatrix} 9 & 6 \\ -3 & 12 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 8 & -4 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ -11 & 16 \end{pmatrix}$

(b) $\det \mathbf{A} = 12+2 = 14$, $\mathbf{A}^{-1} = \frac{1}{14}\begin{pmatrix} 4 & -2 \\ 1 & 3 \end{pmatrix}$

(c) $\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 8 \\ 2 \end{pmatrix} = \frac{1}{14}\begin{pmatrix} 28 \\ 14 \end{pmatrix}$, so $x = 2$, $y = 1$

(d) $\mathbf{N} = 14\mathbf{A}^{-1} = \begin{pmatrix} 4 & -2 \\ 1 & 3 \end{pmatrix}$